

ELEC5305 Project Proposal

Project Title: STFT-Based Speech Enhancement Using Wiener Filtering in Noisy Environments

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GitHub Repository:

<https://github.com/JennyWu55/elec5305-project-550203623>

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<https://jennywu55.github.io/elec5305-project-550203623/>

Project Overview

Speech recordings are often affected by environmental background noise, such as traffic, air-conditioning systems, and surrounding conversations. This noise can reduce speech intelligibility and degrade the quality of recorded audio, which may also affect applications such as voice communication and automatic speech processing. The aim of this project is to investigate a signal-processing approach for enhancing noisy speech using the Short-Time Fourier Transform (STFT) and Wiener filtering. The noisy speech signal will first be transformed into a time-frequency representation using the STFT. A Wiener filter will then be applied to suppress estimated noise components while preserving useful speech information. Finally, the enhanced speech signal will be reconstructed using the inverse STFT. The effectiveness of the method will be evaluated under different noise levels.

Background and Motivation

Speech enhancement is an important area of digital signal processing because speech signals recorded in real environments are rarely free from noise [1], [2]. Background noise can distort both the time-domain waveform and the frequency content of speech, making it more difficult for listeners or speech-processing systems to recover useful information.

One common approach is to analyse speech in short time frames because speech is non-stationary over long periods but can be treated as approximately stationary within short intervals.

The Short-Time Fourier Transform provides a convenient way to represent these short-time speech segments in the time-frequency domain. Once the signal is represented in this form, noise reduction can be performed separately across different frequency components. Wiener filtering is a classical signal-processing method that estimates an appropriate gain for reducing noise while attempting to minimise distortion of the desired signal [1]. More recent speech enhancement research has also explored data-driven and deep-learning approaches to improve noise suppression performance under challenging acoustic conditions [2]-[4].

This topic was selected because it combines several core signal-processing concepts, including windowing, frequency analysis, STFT processing, filtering, and signal reconstruction. It is also suitable for implementation and evaluation in MATLAB and provides clear quantitative and visual results through SNR measurements, waveforms, and spectrograms.

Proposed Methodology

The project will be implemented primarily in MATLAB using publicly available clean speech recordings from the LibriSpeech corpus and environmental noise recordings from the DEMAND database. The clean speech signals will first be mixed with background noise at several input signal-to-noise ratio (SNR) levels, such as -5 dB, 0 dB, 5 dB, and 10 dB. This will allow the performance of the enhancement algorithm to be evaluated under different noise conditions.

The noisy speech signal will be divided into short overlapping frames and multiplied by an analysis window. The Short-Time Fourier Transform (STFT) will then be applied to convert each frame from the time domain

into a time-frequency representation. A noise power spectrum will be estimated using noise-dominant or low-energy frames of the noisy recording, and this estimate will be used to calculate a Wiener gain for each time-frequency component. The Wiener gain will suppress components that are dominated by noise while retaining components that are more likely to contain useful speech information.

After filtering, the enhanced time-frequency representation will be converted back to the time domain using the inverse STFT and overlap-add reconstruction. The resulting enhanced speech will then be compared with the original noisy speech.

Performance will be evaluated using input and output SNR, SNR improvement, waveform comparisons, and spectrograms. Audio examples will also be included to allow a subjective comparison between the noisy and enhanced speech. If time permits, spectral subtraction may be implemented as a baseline method for comparison with Wiener filtering.

Expected Outcomes

The expected outcome of this project is a MATLAB-based speech enhancement prototype capable of reducing background noise from noisy speech recordings using STFT-based Wiener filtering. The system is expected to produce enhanced speech signals with improved signal-to-noise ratio while preserving the main speech components.

The effectiveness of the proposed method will be demonstrated through quantitative and visual comparisons between the noisy and enhanced signals. These will include input and output SNR measurements, SNR improvement, time-domain waveforms, and spectrograms. Audio examples of clean, noisy, and enhanced speech will also be provided for subjective comparison.

The final MATLAB code, project documentation, selected audio examples,

figures, and evaluation results will be organised and published in the project GitHub repository.

Timeline

Weeks 1-2: Finalise the project topic, define the project scope, and set up the GitHub repository and project page.

Weeks 3-5: Conduct a literature review on speech enhancement, STFT processing, and Wiener filtering. Select suitable clean speech and environmental noise recordings for testing.

Weeks 6-7: Implement audio loading, signal normalisation, noise mixing, framing, windowing, and STFT analysis in MATLAB.

Weeks 8-9: Implement Wiener filtering, noise spectrum estimation, inverse STFT, and signal reconstruction. Perform initial testing under different noise conditions.

Weeks 10-11: Evaluate the system using input and output SNR, SNR improvement, waveform comparisons, spectrograms, and listening tests. Refine the implementation if necessary.

Weeks 12-13: Finalise the experimental results, complete the final report, organise the MATLAB code and audio examples, and update the GitHub documentation and project demonstration.

References

- [1] C. Plapous, C. Marro, and P. Scalart, “Improved signal-to-noise ratio estimation for speech enhancement,” *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 14, no. 6, pp. 2098 – 2108, Nov. 2006.
- [2] Y. Xu, J. Du, L.-R. Dai, and C.-H. Lee, “A regression approach to speech enhancement based on deep neural networks,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 23, no. 1, pp. 7 – 19, Jan. 2015.
- [3] D. Wang and J. Chen, “Supervised speech separation based on deep learning: An overview,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 26, no. 10, pp. 1702 – 1726, Oct. 2018.
- [4] A. Pandey and D. Wang, “A new framework for CNN-based speech

enhancement in the time domain,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 27, no. 7, pp. 1179 – 1188, Jul. 2019.